Related Work: Classifying exoplanet candidates with convolutional neural networks: application to the Next Generation Transit Survey

Key Innovation

CNN trained on mix of real NGTS observations, unlabeled data, and simulated transits

Key Takeaway

Extra features (radius, period) help distinguish planets from eclipsing binaries

Architecture

- Dual-view CNN (global + local light curve views)
- Additional stellar features: radius and orbital period
- Trained with Adam optimizer and early stopping

Results

- 96% accuracy, 99.3% AUC
- Recovered 13/14 confirmed planets

Chaushev, A., Raynard, L., Goad, M. R., Eigmüller, P., Armstrong, D. J., Briegal, J. T., Burleigh, M. R., Casewell, S. L., Gill, S., Jenkins, J. S., Nielsen, L. D., Watson, C. A., West, R. G., Wheatley, P. J., Udry, S., & Vines, J. I. (2019). Classifying exoplanet candidates with convolutional neural networks: Application to the Next Generation Transit Survey. Monthly Notices of the Royal Astronomical Society, 488(4), 5232–5250. https://doi.org/10.1093/mnras/stz2058

Related Work: Flare Statistics for Young Stars from a Convolutional Neural Network Analysis of TESS Data

Key Innovation

CNN learns flare shapes from 200-cadence segments centered on events

Key Takeaway

Segmented light curves centered on events can train CNNs for transient detection

Architecture

- 1D CNN with global max pooling and dropout
- Binary classification: flare vs. non-flare
- 10-model ensemble for stability

Results

- AUC of 0.993, F1 above 0.98
- Over 17,000 flares found in 3,200 young stars

Feinstein, A. D., Montet, B. T., Ansdell, M., Nord, B., Bean, J. L., Günther, M. N., Gully-Santiago, M. A., & Schlieder, J. E. (2020). Flare Statistics for Young Stars from a Convolutional Neural Network Analysis of TESS Data. The Astronomical Journal, 160(5), 219. https://doi.org/10.3847/1538-3881/abac0a

Related Work: The GPU Phase Folding and Deep Learning Method for Detecting Exoplanet Transits

Key Innovation
GPU-parallelized phase folding enables exhaustive period searches

Key Takeaway
GPU acceleration makes exhaustive period searches practical for large datasets

Method

- Tests 100,000 trial periods in 5 seconds (RTX 3090 Ti)
- 19-layer CNN with Circular Convolution
- Trained on 2 Million Synthetic Light Curves

Results

- 97% accuracy on detecting ultra-short-period planets
- 3 orders faster than BLS

Wang, K., Ge, J., Willis, K., Wang, K., & Zhao, Y. (2024). The GPU Phase Folding and Deep Learning Method for Detecting Exoplanet Transits. Monthly Notices of the Royal Astronomical Society. Advance Access publication 2024 January 23. https://academic.oup.com/mnras/article/528/3/4053/7585892

Progress Since Proposal

- Dataset Downloader
- Preprocessing pipeline baseline
- Dual-branch CNN architecture baseline
- Hyperparameter optimization with Optuna
- Training/Testing baseline

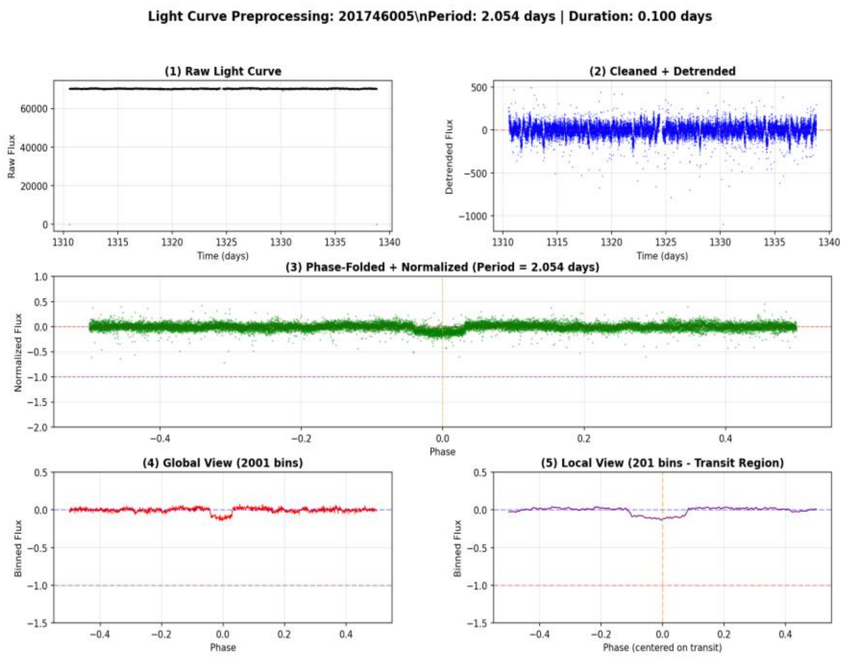
Methodology: Dual-Branch CNN

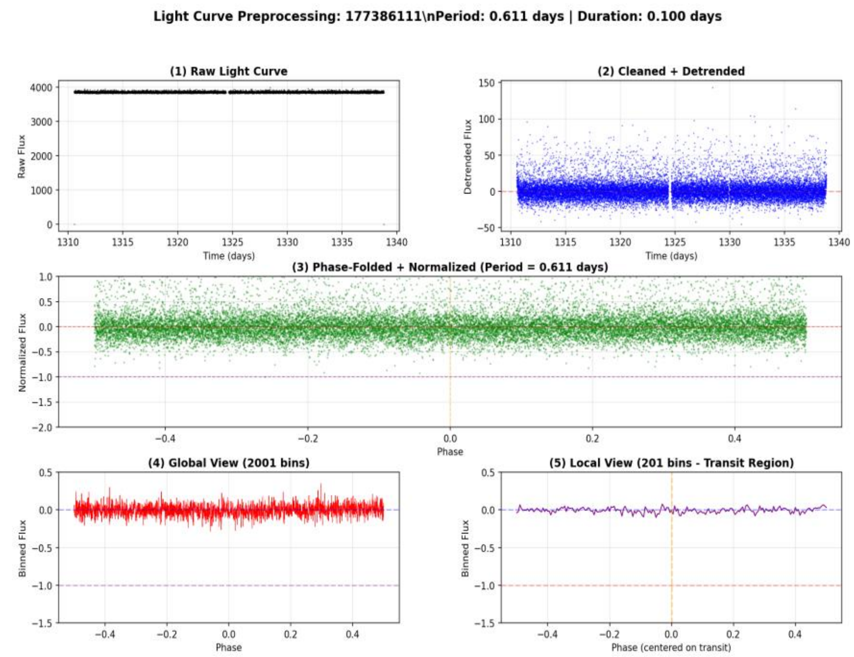
Data

- 2821 samples (70/15/15 train/val/test split)

Preprocessing

- 5σ clipping, Median Filter, BLS Period Estimation
- Phase folding, Dual-view creation (Global 2001 + Local 201)





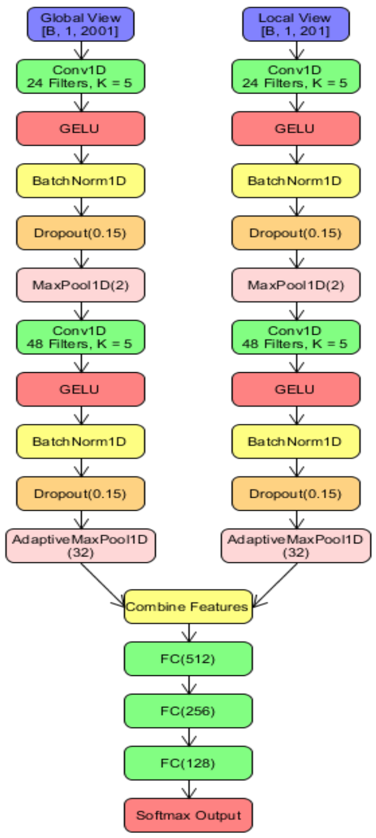
Methodology: Dual-Branch CNN (Cont.)

Model Architecture

- Dual-branch CNN: 2 Conv blocks/branch (24 filters, k=5)
- GELU + BatchNorm, Concatenate, FC: 512, 256, 128
- 25 Layers Total

Training

- Batch Sampling: 32
- AdamW Optimizer (lr=1e-3) + Class-weighted loss
- On-the-fly data augmentation (Gaussian Noise + Flux Scaling + Phase Shifts)
- Early Stopping



Experimental Results

Dataset & Training

TESS Lilith 4 Dataset

2821 samples total

- Training: 1974 (70%)
- Validation: 423 (15%)
- Test: 424 (15%)

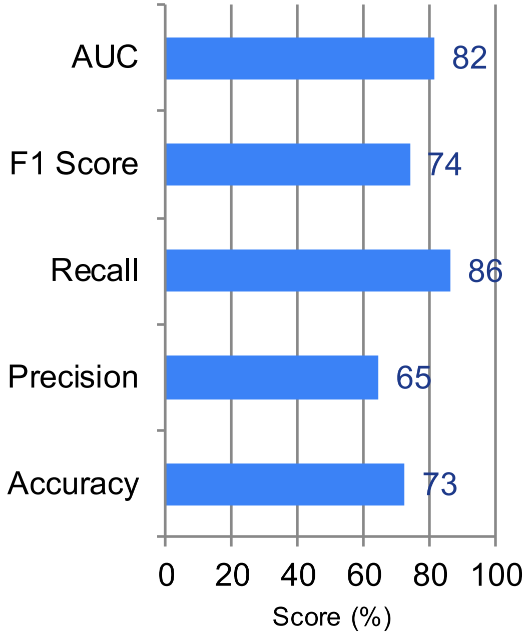
Class Distribution

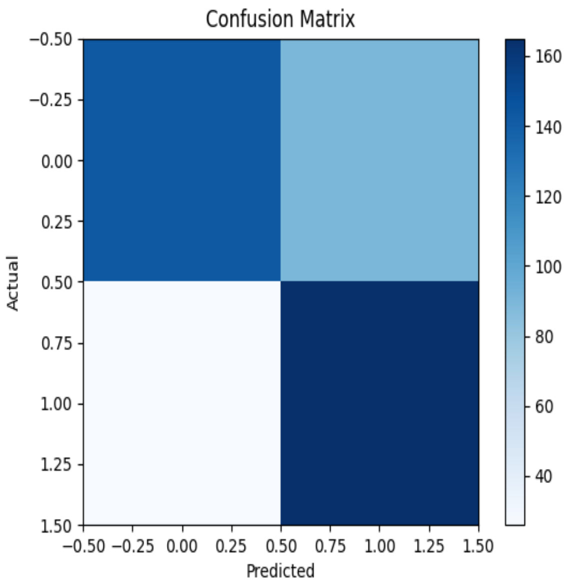
- Planets: 52.4%
- Non-planets: 47.6%

Key Result

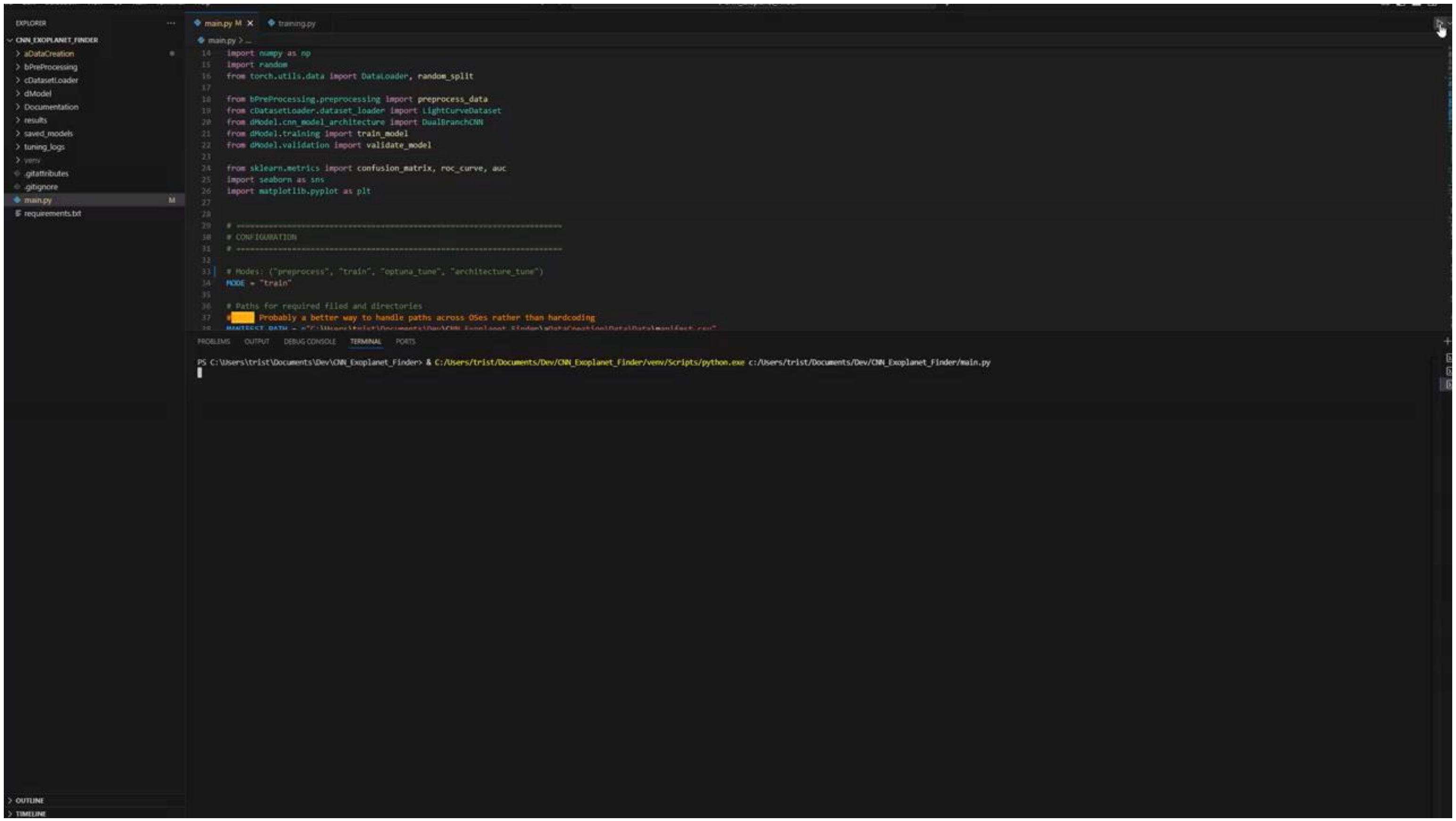
High recall for discovery

Test Set Performance





Demo


EXPLORER
CNN_EXOPLANET_FINDER
aDataCreation
bPreProcessing
cDatasetLoader
dModel
Documentation
results
saved_models
tuning_logs
venv
.gitattributes
.gitignore
main.py
requirements.txt
OUTLINE
TIMELINE
main.py M
training.py
import numpy as np
import random
from torch.utils.data import DataLoader, random_split
from bPreProcessing.preprocessing import preprocess_data
from cDatasetLoader.dataset_loader import LightCurveDataset
from dModel.cnn_model_architecture import DualBranchCNN
from dModel.training import train_model
from dModel.validation import validate_model
from sklearn.metrics import confusion_matrix, roc_curve, auc
import seaborn as sns
import matplotlib.pyplot as plt
# CONFIGURATION
# Modes: ("preprocess", "train", "optuna_tune", "architecture_tune")
MODE = "train"
# Paths for required filed and directories
Probably a better way to handle paths across OSes rather than hardcoding
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\trist\Documents\Dev\CNN_Exoplanet_Finder> & C:/Users/trist/Documents/Dev/CNN_Exoplanet_Finder/venv/Scripts/python.exe c:/Users/trist/Documents/Dev/CNN_Exoplanet_Finder/main.py

What's Next?

Short Term

- Expand the training dataset beyond Sector 1
- Implement k-fold validation
- Test on K2 and Kepler data

Medium Term

- Integrate centroid information
- Train on mixed dataset (generalization)
- Find the best parameters for 10 models
- Build an ensemble model

Long Term

- Explore GPU Phase Fold Acceleration
- Add auxiliary features

Final Deliverables

- Research Paper
- CNN Model
- Code

Transformers

New research papers

- Chen, T.; Ren, X.; Lai, J.; Tan, H.; Liu, F.; and Chan, W. K. V. 2025. Toward Transformer-compatible multivariate time series learning via visibility graph-based structural encoding. Knowledge-Based Systems, 329: 114389.
- Pan, J.-S.; Ting, Y.-S.; and Yu, J. 2024. Astroconformer: The prospects of analysing stellar light curves with transformer-based deep learning models. Monthly Notices of the Royal Astronomical Society, 528(4): 5890–5903.
- Wen, Q.; Zhou, T.; Zhang, C.; Chen, W.; Ma, Z.; Yan, J.; and Sun, L. 2023. Transformers in Time Series: A Survey. 32nd International Joint Conference on Artificial Intelligence (IJCAI 2023). arXiv:2202.07125.

Key Innovation

Key Takeaway

Chen, T.; Ren, X.; Lai, J.; Tan, H.; Liu, F.; and Chan, W. K. V. 2025. Toward Transformer-compatible multivariate time series learning via visibility graph-based structural encoding. Knowledge-Based Systems, 329: 114389.

Key Innovation

Key Takeaway

Pan, J.-S.; Ting, Y.-S.; and Yu, J. 2024. Astroconformer: The prospects of analysing stellar light curves with transformer-based deep learning models. Monthly Notices of the Royal Astronomical Society, 528(4): 5890–5903.

Key Innovation

Key Takeaway

Wen, Q.; Zhou, T.; Zhang, C.; Chen, W.; Ma, Z.; Yan, J.; and Sun, L. 2023. Transformers in Time Series: A Survey. 32nd International Joint Conference on Artificial Intelligence (IJCAI 2023). arXiv:2202.07125.

Progress Since Proposal

Methodology: Transformer

Data

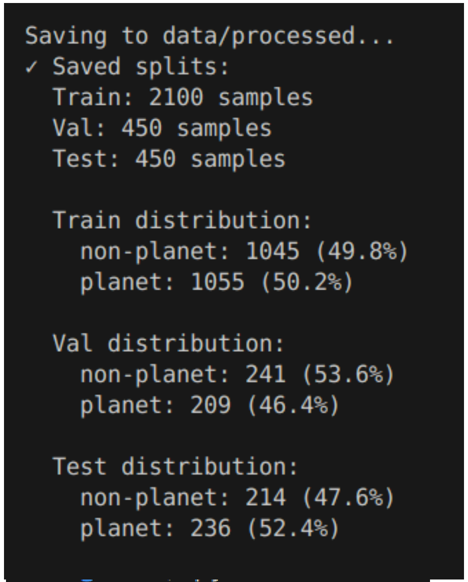
- 3000 samples from TESS Lilith 4 Dataset, sector 1, with balanced distribution (70/15/15 train/val/test split)

Preprocessing Steps

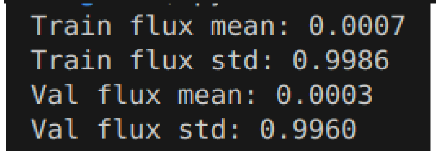
- **Sigma-clipping –** removes outlier flux points
- **Savitzky-Golay filter –** removes low frequency trends
- **Gap Interpolation –** fills in short gaps with linear estimation
- **Z-score Normalization –** sets mean to zero and std dev to one
- **Segmentation –** standardizes segments to 27 day periods
- **Augmentation**
    - Gaussian noise
    - Time Shift
    - Amplitude scaling
    - Time warping

Hyperparameter Optimization

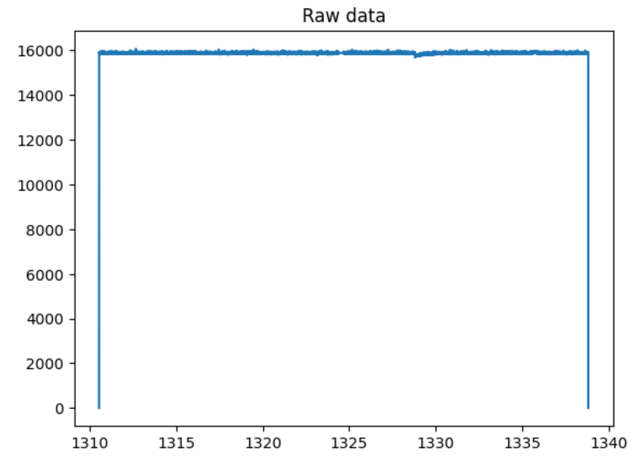
- 50 trials with 15 epochs using Optuna

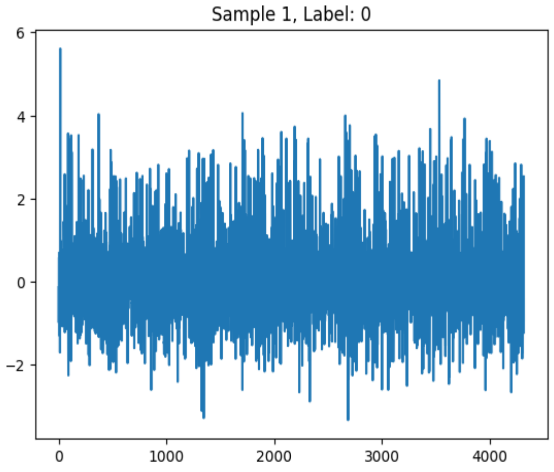


```
Saving to data/processed...
✓ Saved splits:
  Train: 2100 samples
  Val: 450 samples
  Test: 450 samples

  Train distribution:
    non-planet: 1045 (49.8%)
    planet: 1055 (50.2%)

  Val distribution:
    non-planet: 241 (53.6%)
    planet: 209 (46.4%)

  Test distribution:
    non-planet: 214 (47.6%)
    planet: 236 (52.4%)
```



```
Train flux mean: 0.0007
Train flux std: 0.9986
Val flux mean: 0.0003
Val flux std: 0.9960
```





Methodology: Transformer

Model Architecture

- **Convolutional Embedding –** 1D CNN for time-series pattern extraction and data embedding
- **Temporal Positional Encoding –** Positional encoding using normalized time stamps
- **Transformer Encoder Blocks –** multi-head self-attention with feed-forward, normalization, and dropout for regularization
- **Global Pooling and Classification –** summarizes features and determines output class

Training

- Linear learning rate warmup (5 epochs) then cosine annealing scheduler
- Label smoothing (1.1%)
- Gradient clipping
- Early Stopping
- Checkpoints

```
Class weights: tensor([1.0048, 0.9952])
Using label smoothing: 0.011383962924475872
Using warmup scheduler: 5 warmup epochs
Using gradient clipping: 1.951167441590346
```

Experimental Results

Dataset & Training

Training Data

2100 original samples (balanced set)

x 2 augmentation factor

4200 samples of training data

Test Data

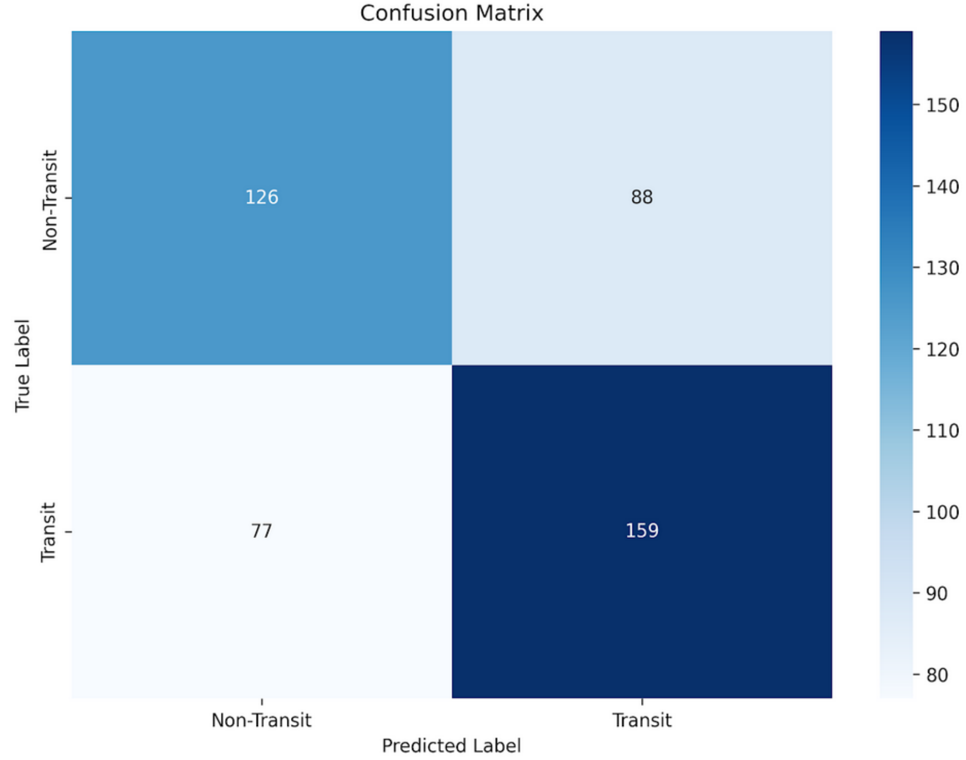
450 samples

- Planets: 236
- Non-planets: 214

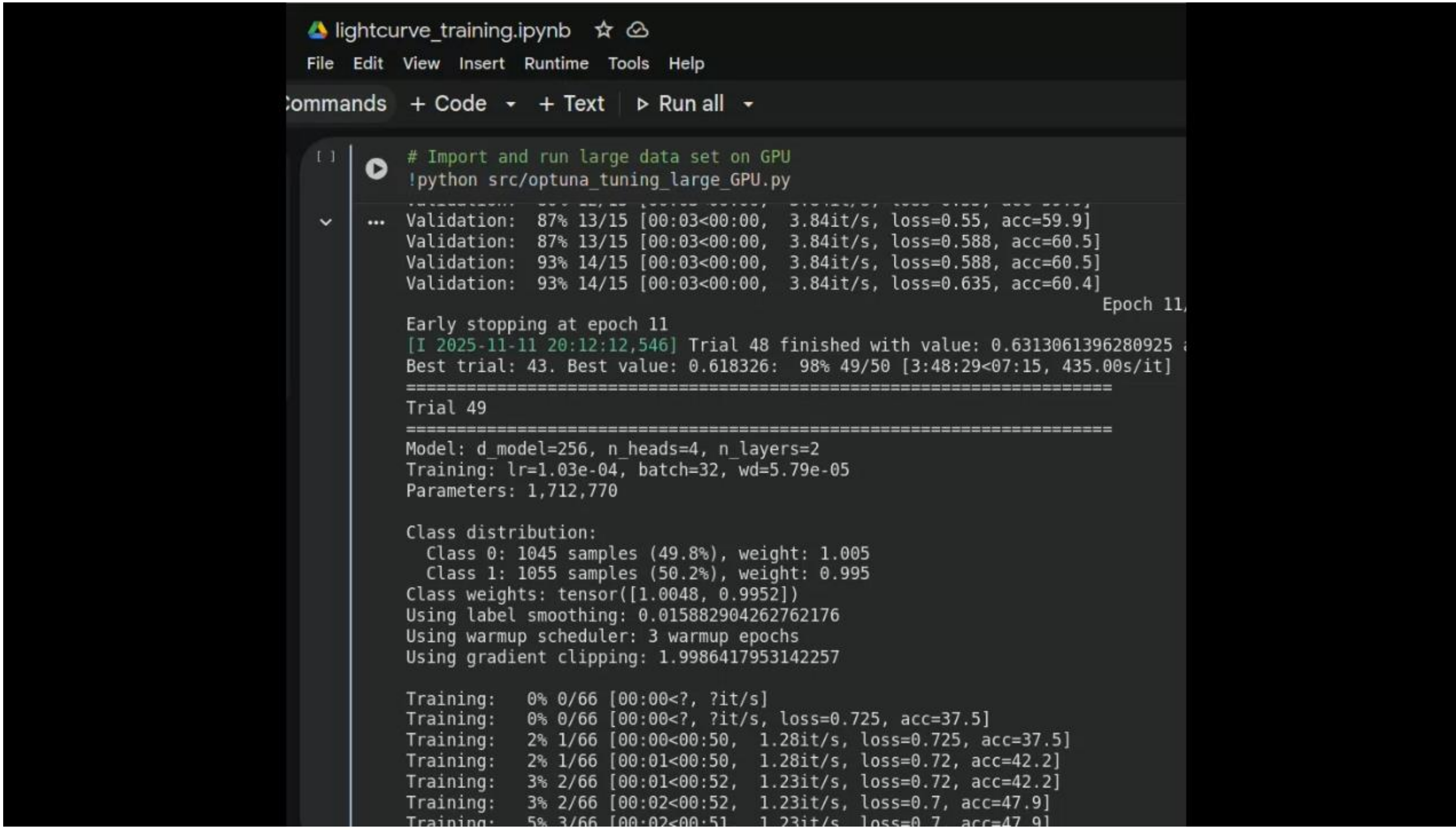
Result Metrics

Accuracy	0.6333
Precision	0.6437
Recall	0.6737
f1_score	0.6584
ROC_AUC	0.6766

Test Set Performance



Demo


lightcurve_training.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Import and run large data set on GPU
!python src/optuna_tuning_large_GPU.py
Validation: 87% 13/15 [00:03<00:00, 3.84it/s, loss=0.55, acc=59.9]
Validation: 87% 13/15 [00:03<00:00, 3.84it/s, loss=0.588, acc=60.5]
Validation: 93% 14/15 [00:03<00:00, 3.84it/s, loss=0.588, acc=60.5]
Validation: 93% 14/15 [00:03<00:00, 3.84it/s, loss=0.635, acc=60.4]
Epoch 11,
Early stopping at epoch 11
[I 2025-11-11 20:12:12,546] Trial 48 finished with value: 0.6313061396280925
Best trial: 43. Best value: 0.618326: 98% 49/50 [3:48:29<07:15, 435.00s/it]
==
Trial 49
==
Model: d_model=256, n_heads=4, n_layers=2
Training: lr=1.03e-04, batch=32, wd=5.79e-05
Parameters: 1,712,770
Class distribution:
Class 0: 1045 samples (49.8%), weight: 1.005
Class 1: 1055 samples (50.2%), weight: 0.995
Class weights: tensor([1.0048, 0.9952])
Using label smoothing: 0.015882904262762176
Using warmup scheduler: 3 warmup epochs
Using gradient clipping: 1.9986417953142257
Training: 0% 0/66 [00:00<?, ?it/s]
Training: 0% 0/66 [00:00<?, ?it/s, loss=0.725, acc=37.5]
Training: 2% 1/66 [00:00<00:50, 1.28it/s, loss=0.725, acc=37.5]
Training: 2% 1/66 [00:01<00:50, 1.28it/s, loss=0.72, acc=42.2]
Training: 3% 2/66 [00:01<00:52, 1.23it/s, loss=0.72, acc=42.2]
Training: 3% 2/66 [00:02<00:52, 1.23it/s, loss=0.7, acc=47.9]

What's Next?

Short Term

Medium Term

Long Term

Final Deliverables

- Research Paper
- Transformer Model
- Code

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